

Customer Churn Analysis

The first use-case that we are to look into is the Customer churn. What customer churn reflects is the rate at which the customer leaves the organisation. This use case is critical for not only the travel business but for any organisation that works on subscriptions especially travelling lodging and bookings.

Here we have used a synthetic data that was generated using a python code which provided us with the relevant information that we can work on. Features in the data set are as follows

- CustomerID
- Age
- ServiceFrequency
- SubscriptionLength
- SatisfactionScore
- AverageSpending
- LastActivityDays
- LoyaltyProgram
- Complaints
- Region
- Churn
- LastActivityDate
- LastActivityMonth

The features mentioned above provide us with an accurate starting point to move ahead with our analysis. We have the data downloaded to csv file which includes total records of 10000. Let's start by first providing brief descriptions of each of the feature that we have to understand what we have to start with.

- **CustomerID**
 - *Type:* Unique Identifier
 - *Description:* A unique numeric identifier for each customer in the dataset.
 - *Characteristics:* Non-repeating and categorical. Not used for analysis but essential for tracking individual records.
- **Age**
 - *Type:* Numerical
 - *Description:* Represents the age of the customer.
 - *Characteristics:* Continuous and distributed across a range; may exhibit patterns like clustering in specific age groups.
- **ServiceFrequency**
 - *Type:* Numerical

- *Description*: The frequency of service usage over a given period.
 - *Characteristics*: Positive integers indicating engagement; values may cluster based on customer behavior.
- **SubscriptionLength**
 - *Type*: Numerical
 - *Description*: Total duration of the customer's subscription in months or days.
 - *Characteristics*: Continuous and typically skewed, with longer durations for loyal customers.
- **SatisfactionScore**
 - *Type*: Numerical
 - *Description*: A rating on a scale (e.g., 1-10) reflecting customer satisfaction.
 - *Characteristics*: Discrete and bounded; lower scores suggest dissatisfaction.
- **AverageSpending**
 - *Type*: Numerical (Float)
 - *Description*: The average monetary amount spent by the customer during their subscription.
 - *Characteristics*: Continuous and often right-skewed, with higher values indicating premium customers.
- **LastActivityDays**
 - *Type*: Numerical
 - *Description*: Days since the customer's last activity or engagement.
 - *Characteristics*: Continuous and temporal; higher values indicate lower engagement.
- **LoyaltyProgram**
 - *Type*: Categorical (Binary)
 - *Description*: Indicates whether the customer is enrolled in a loyalty program (1 = Enrolled, 0 = Not Enrolled).
 - *Characteristics*: Dichotomous and directly linked to customer retention strategies.
- **Complaints**
 - *Type*: Numerical
 - *Description*: Number of complaints lodged by the customer.
 - *Characteristics*: Discrete, with higher values suggesting dissatisfaction.
- **Region**
 - *Type*: Categorical (Nominal)
 - *Description*: Geographic location of the customer (e.g., Urban, Suburban, Rural).
 - *Characteristics*: Non-ordinal categories reflecting demographic diversity.
- **Churn**
 - *Type*: Categorical (Binary)
 - *Description*: Target variable indicating if a customer has churned (1 = Churned, 0 = Not Churned).
 - *Characteristics*: Binary label used for churn analysis.
- **LastActivityDate**
 - *Type*: Temporal
 - *Description*: Timestamp of the last recorded customer activity.

- *Characteristics*: Used to derive temporal features like last activity month or days since activity.
- **LastActivityMonth**
 - *Type*: Temporal (Derived)
 - *Description*: The month in which the customer was last active.
 - *Characteristics*: Captures seasonal or monthly trends in activity.

Data Analysis

We will start by understanding the basics of our data. Here our crucial information is related to customer, which might include its age, frequency of the service and their average spending. **Table 1.1** shows the Average, Median and Standard Deviation for each of the mentioned features.

Table 1.1

	Average	Median	Standard Deviation
Age	43.5394	43.0	14.911636
ServiceFrequency	10.1614	10.0	5.497179
AverageSpending	2748.7204	2741.5	1299.596426

The **figure 1** below provides a visual representation of the distribution of Age, ServiceFrequency and AverageSpending in the data set.

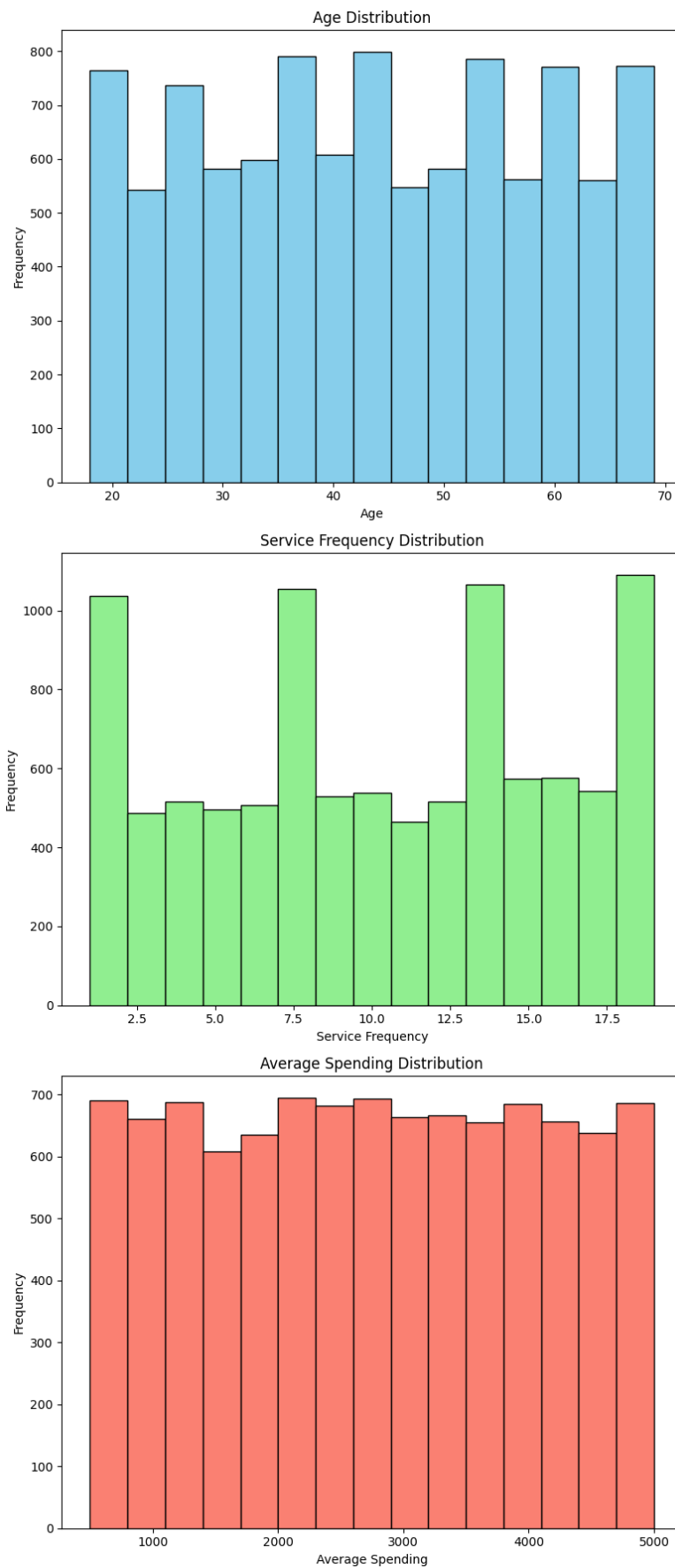


Figure 1: General distribution for key features

To have a clearer understanding of the relation between the variables and the churn we will step into correlation analysis and visualise it. The table **Table 2** brings correlation into visualization that depicts the relation of the limited set of features that could be crucial for calculation of the churn.

Table 2: Correlation values for key features

	LastActivityDays	SatisfactionScore	Churn
LastActivityDays	1.000000	-0.002029	-0.011151
SatisfactionScore	-0.002029	1.000000	-0.006654
Churn	-0.011151	-0.006654	1.000000

To bring a clear picture into context to understand the relationship figure 2 is useful.

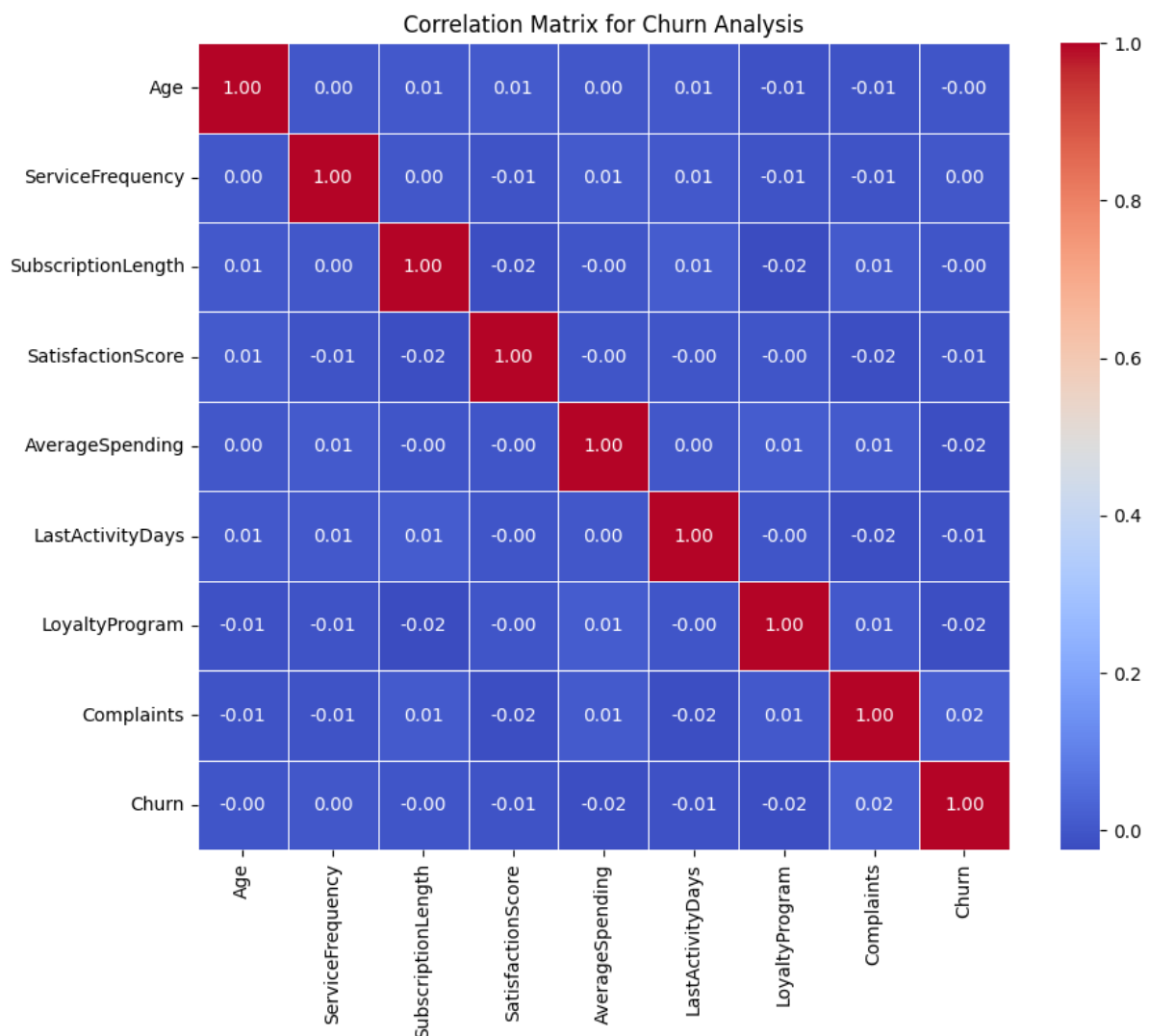


Figure 2: Correlation Matrix for Churn Analysis

The graph provides a clear understanding of the relationship between different feature sets and churn. For instance, the correlation between Complaints and churn is relatively weak, with a value of 0.02, while customers enrolled in a loyalty program show a negative correlation with the churn rate.

Similarly, we are able to analyze relationships within feature sets. For example, the correlation between Complaints and Satisfaction Score shows a negative effect, with a correlation coefficient of -0.02.

It is important to note that the relationships indicated by the correlation values do not appear to be very strong. This is primarily due to the fact that the dataset we have been working with is synthetic and may not closely reflect real-world scenarios.

Now, let's examine churn by Region and Age. For this analysis, it is helpful to divide customers into multiple age groups to better understand the trend for each segment. Therefore, we categorized the customers into five distinct age groups.

- 18 - 30
- 31 - 40
- 41 - 50
- 51 - 60
- 61 - 70

The corresponding churn rate for each of the age groups can be seen in the first pie-chart of the **figure 1.3**. It ranges from the lowest value of 18.5% in 41-50 age groups to the highest one that is observed in 51-60yrs of 21.9%.

The second pie-chart depicts the churn rate as per the 3 different regions as Rural, Urban, Suburban. (**Figure 1.3**)

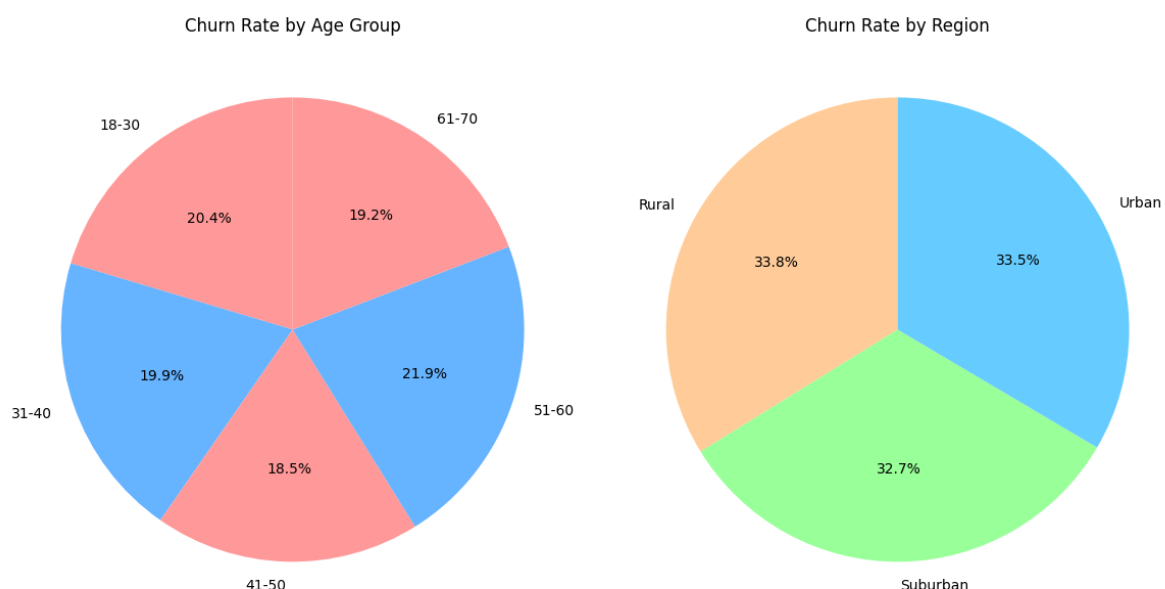


Figure 3: Churn Rate by Age Group & Region

Customer behavior analysis also provides valuable insights into the thinking patterns of customers. For example, features such as Service Frequency, Average Spending, and Last Activity Days offer

significant information about the behaviors they follow. Consider a customer whose last activity was a year ago—there is a high likelihood that this customer may discontinue using the services. However, before making a definitive decision, other factors should be considered, such as the frequency of service usage and the pattern since the customer's first use. While these factors may indicate a higher probability of the customer being at high risk, further analysis can help refine the prediction.

There may be instances where outliers exist, such as a customer experiencing a personal setback or unforeseen circumstances that may result in them leaving the services. However, the probability of such events occurring is relatively low, so these outliers can generally be disregarded in the analysis.

To interpret this behavior pattern, we can identify a threshold for risky behavior, such as spending below the mean value. Customers who fall below this threshold are labeled as "Potential Churn Cases," and those spending close to or below the minimum value are categorized as "High Risk." This approach is valuable for monitoring customers who are most likely to stop using the services. Figure 1.4 presents a table that provides insightful data, enabling informed decision-making by analyzing the trends.

Metric	Churned							Non-Churned						
	Mean	Std Dev	Min	25th Percentile	Median (50%)	75th Percentile	Max	Mean	Std Dev	Min	25th Percentile	Median (50%)	75th Percentile	Max
ServiceFrequency	10.18	5.48	1	5	10	15	19	10.16	5.50	1	5	10	15	19
AverageSpending	2703.15	1296.28	502	1599	2683	3822	4996	2760.28	1300.26	500	1636	2754	3889	4999
LastActivityDays	178.99	106.03	0	87	177	274	364	181.92	105.08	0	91	182	272	364

Figure 4: Matrics to analyse key features

The next visualization helps us analyze the impact of customer satisfaction on churn. To do this, we compared the average Satisfaction Score between churned and non-churned customers and segmented customers by Satisfaction Score to calculate the churn rates for each segment. Based on the values from the graph in Figure 1.5, we observed that non-churned customers tend to have slightly higher satisfaction scores, while churned customers exhibit relatively lower scores.

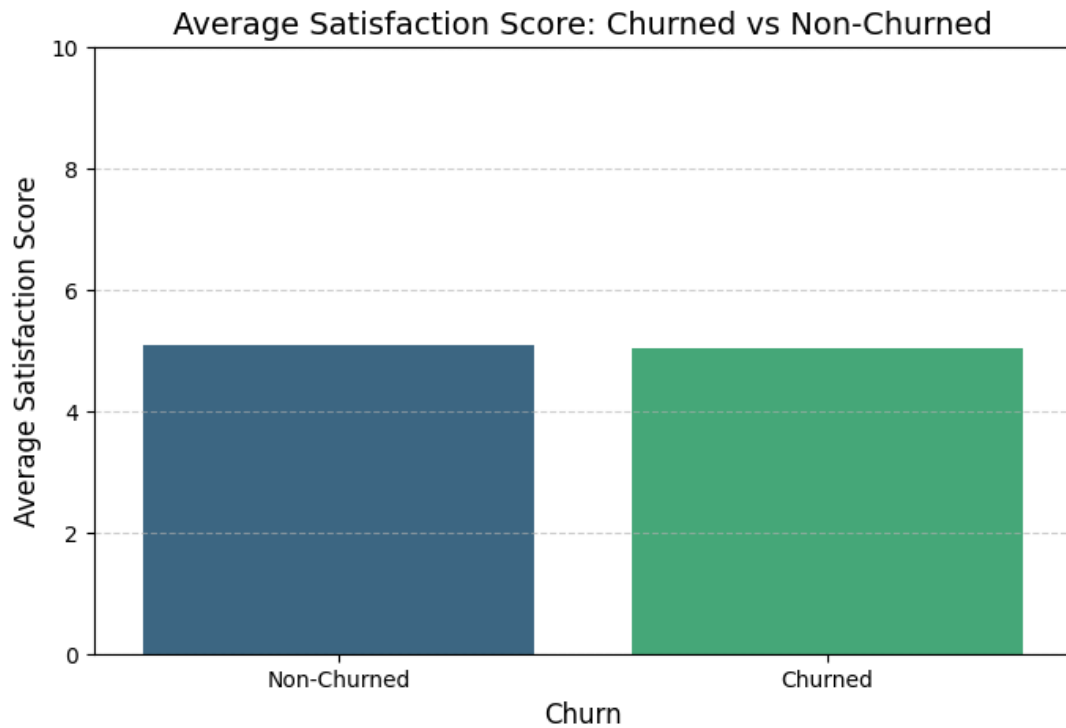


Figure 5: Avg Satisfaction Score (Churned vs Non-Churned)

A similar approach is taken when analyzing the relationship between customer complaints and churn rate. As shown in the bar graph in Figure 1.6, approximately 6,500 customers filed a complaint, while slightly over 3,000 did not. The subsequent graph reveals a slight difference in the churn rate between these two groups. A higher churn rate among customers who filed complaints suggests unresolved issues, highlighting the need to focus on improving the complaint resolution process. This will help reduce the number of customers leaving due to subpar customer service.

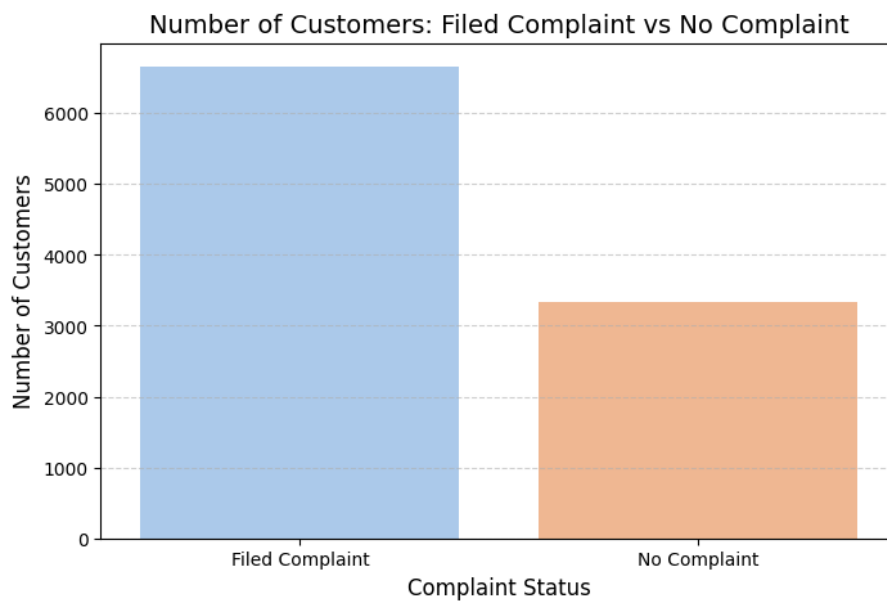


Figure 6: Customer distribution by complaints

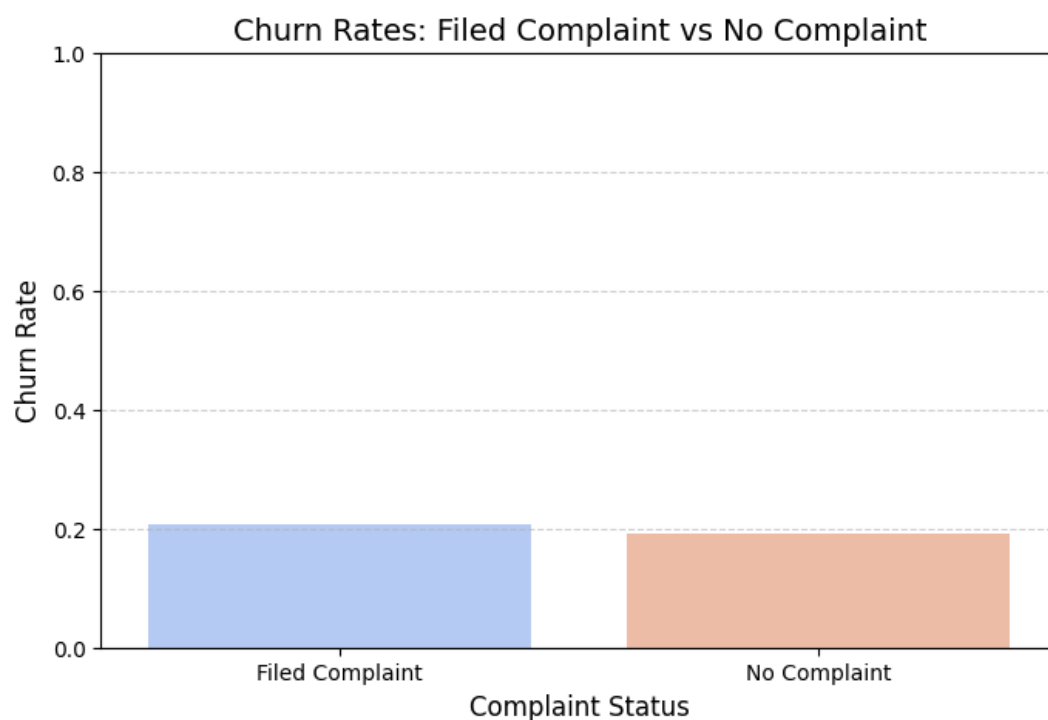


Figure 7: Churn rate (Complaints vs Non Complaint)

Focusing on the effectiveness of the Loyalty Program, we observe a slightly higher churn rate among individuals not participating in the program compared to those who are enrolled. This feature is critical for analyzing the relationships, effects, and correlations between various factors. To further investigate, we proceed with visualizing the relationship between the Loyalty Program and Average Spending, as well as between the Loyalty Program and Satisfaction Score. The resulting graphs clearly highlight the differences, providing valuable insights.

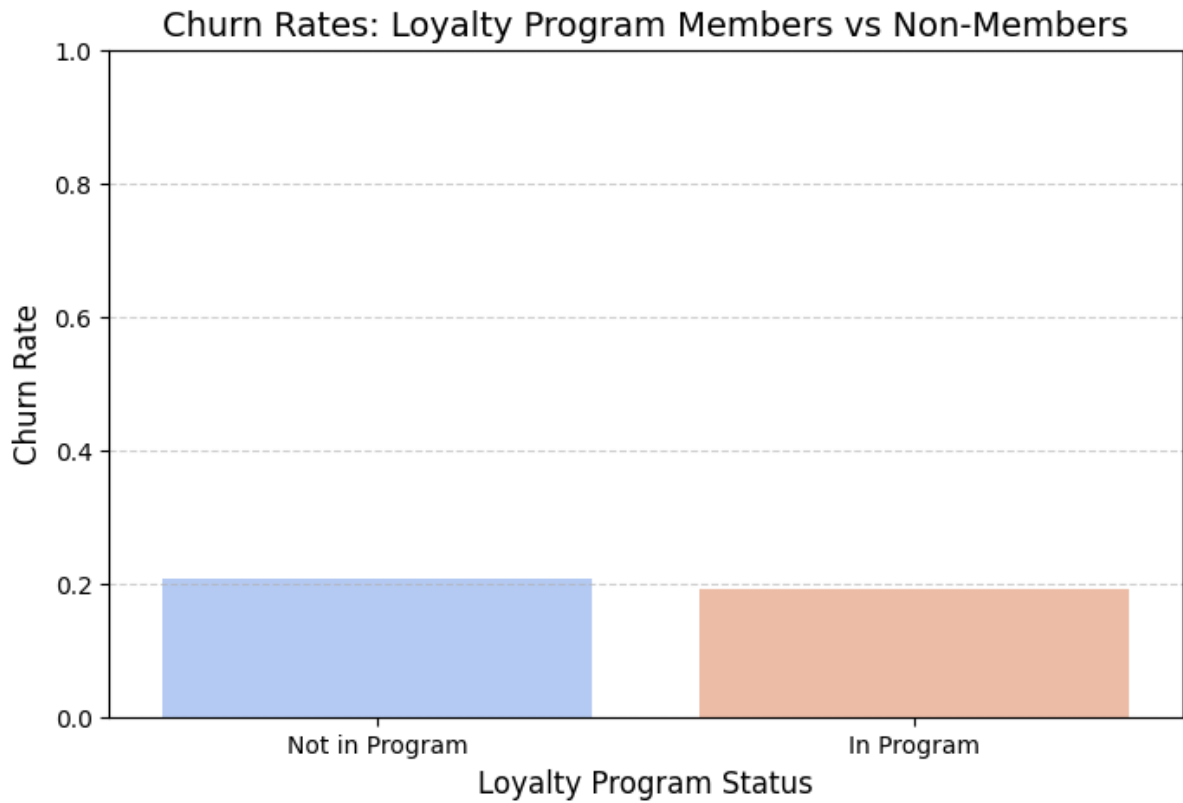


Figure 8: Churn Rate by Loyalty Program

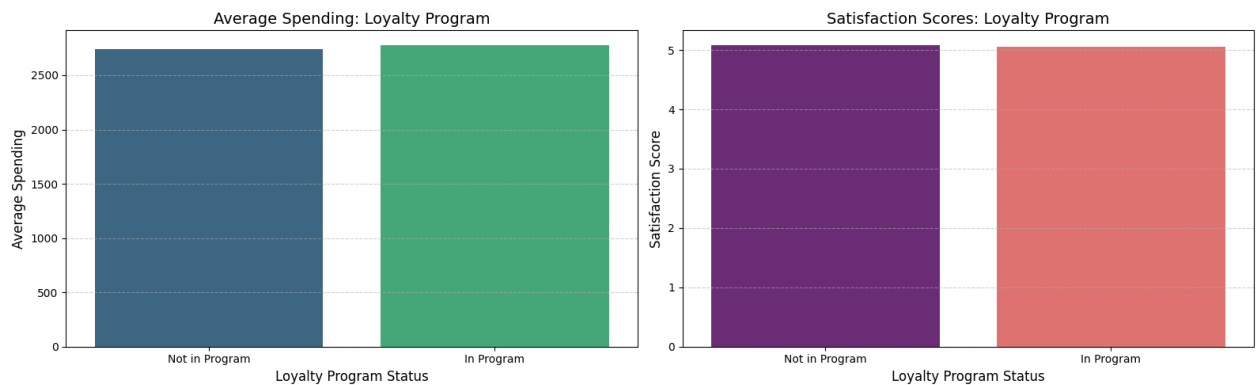


Figure 9: Avg Spending & Satisfaction score vs Loyalty Status

This analysis highlights the importance of investing in loyalty program initiatives. Expanding the program to include more customers can significantly reduce the risk of churn. Additionally, it is crucial to evaluate the program's design to ensure it aligns with customer expectations. If any aspects of the program are causing dissatisfaction or disengagement, restructuring it could prove beneficial in retaining customers and lowering churn rates in the future.

An essential step to uncover further insights is the analysis of declining spending or satisfaction over time, represented by the subscription length of customers. This approach involves comparing **AverageSpending** with **SubscriptionLength** to identify patterns and trends over extended periods. The line plot shown in Figure 1.9 effectively visualizes these relationships, providing a clear depiction of how spending behaviors or satisfaction levels evolve as subscription lengths increase.

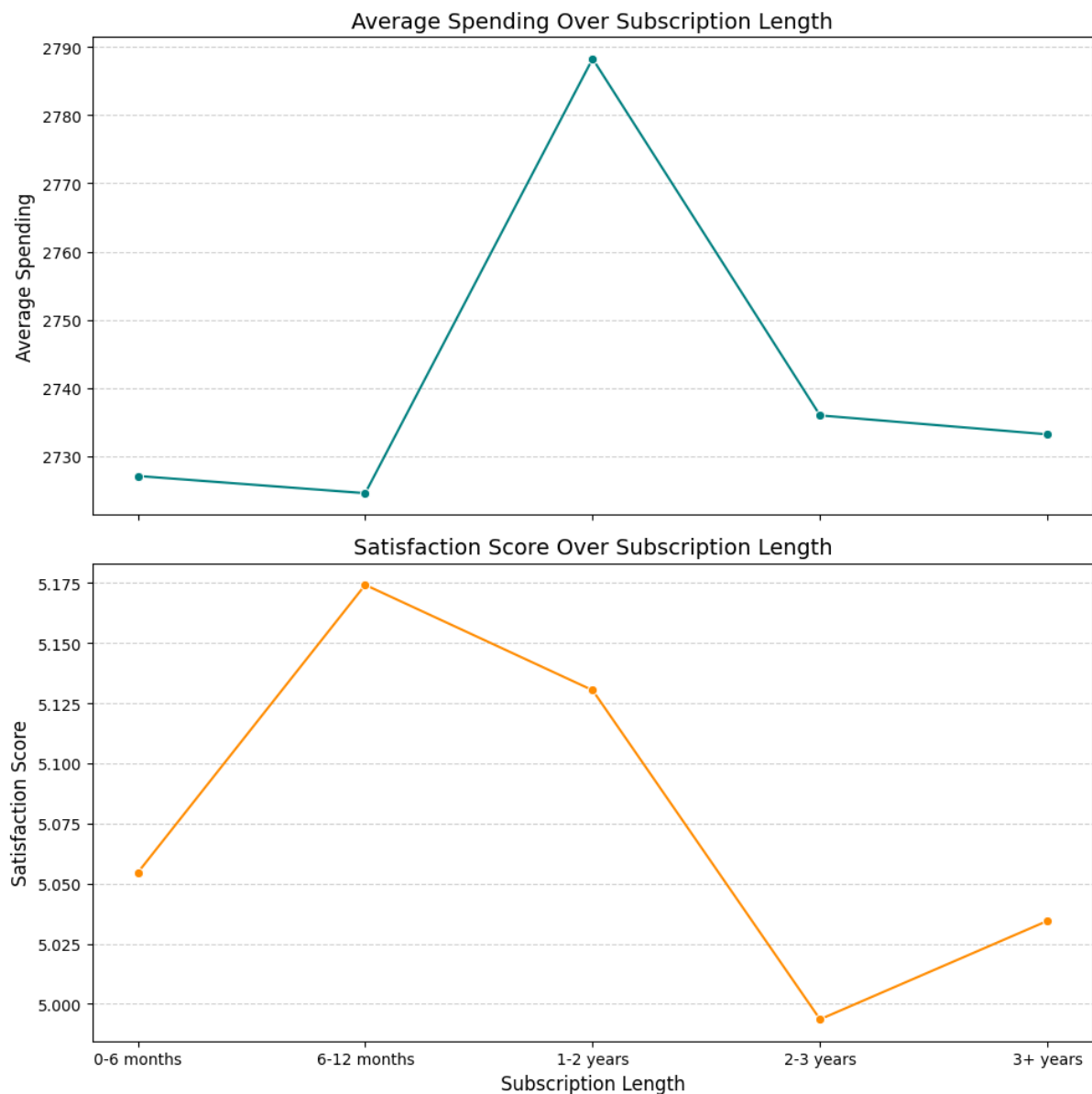


Figure 10: Avg Spent & Satisfaction score vs Subscription Length

A downward trend in spending and satisfaction scores over longer subscription periods highlights a decline in customer engagement. This trend suggests the need for targeted retention strategies aimed at customers in mid-to-long subscription periods who exhibit signs of disengagement. By proactively addressing this issue, we can enhance customer satisfaction and loyalty, mitigating the risk of churn in this segment.

Identifying the correct areas of focus is essential to designing impactful strategies, and making informed decisions is critical to achieving success. Figure 1.10 exemplifies this approach by identifying customers in the "Rural" region with low satisfaction. Customers in rural areas are often at a higher risk of discontinuing services compared to those in urban areas, likely due to differences in spending power and access to resources. To address this, we filtered the dataset for customers in the "Rural" region with a low SatisfactionScore, defining "low satisfaction" as a score below 4. This analysis allows us to visualize the distribution of satisfaction scores among rural customers, providing actionable insights to target retention efforts effectively.

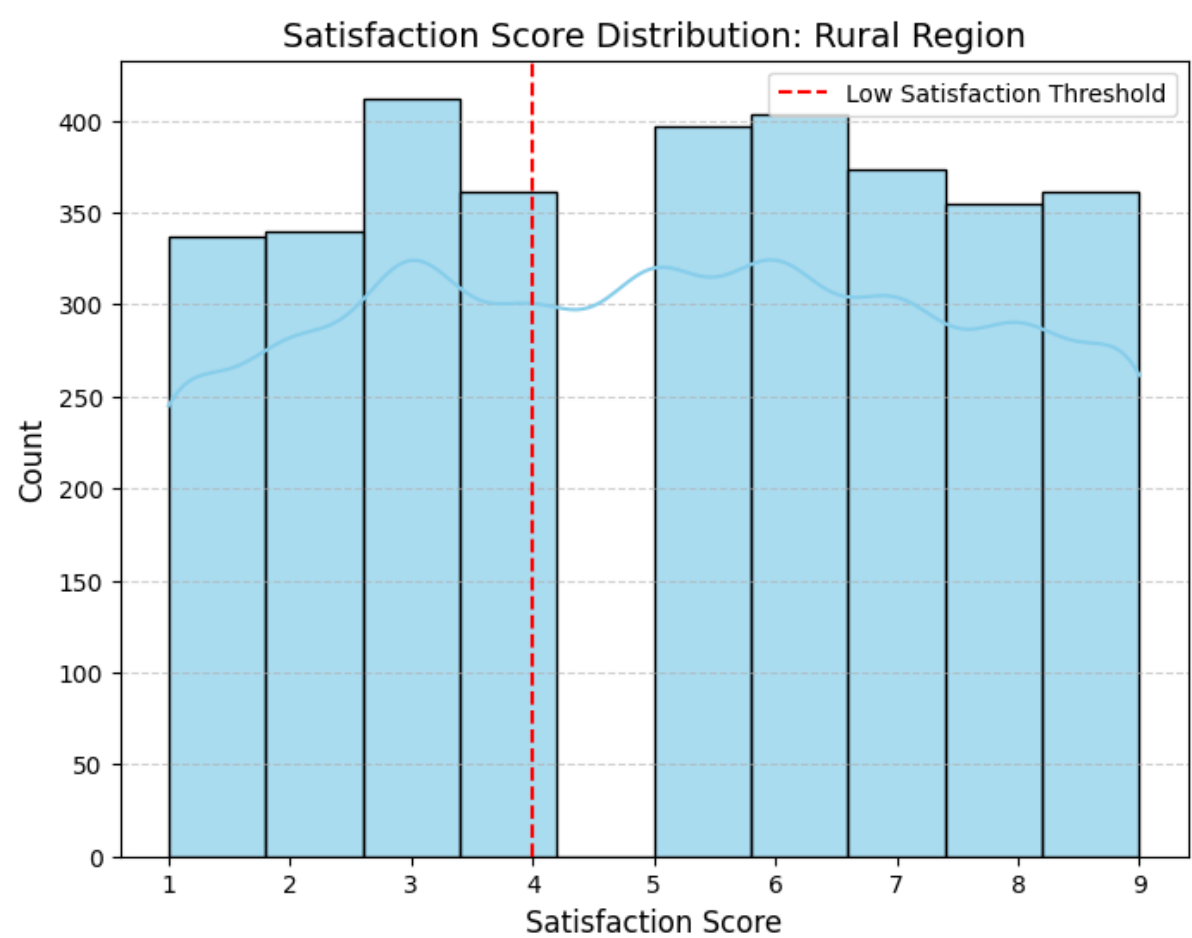


Figure 11: Satisfaction score distribution in Rural area

Churn Prediction Model (XGBoost)

We leveraged the capabilities of a machine learning model to predict customer churn by analyzing key predictors such as Age, ServiceFrequency, SatisfactionScore, LastActivityDays, LoyaltyProgram, and Complaints. Using XGBoost, we trained the model on 70% of the dataset and evaluated its performance on the remaining 30%. Metrics such as accuracy, precision, recall, and F1-score were

used to assess the model's effectiveness. The analysis revealed key drivers of churn, as shown in Figure 1.11. Among these, LastActivityDays emerged as the most significant feature influencing churn prediction, followed by Age, SatisfactionScore, ServiceFrequency, and Complaints. With these insights from the trained model, we can now proactively target high-risk customers and design retention strategies informed by the features that play a critical role in driving churn.

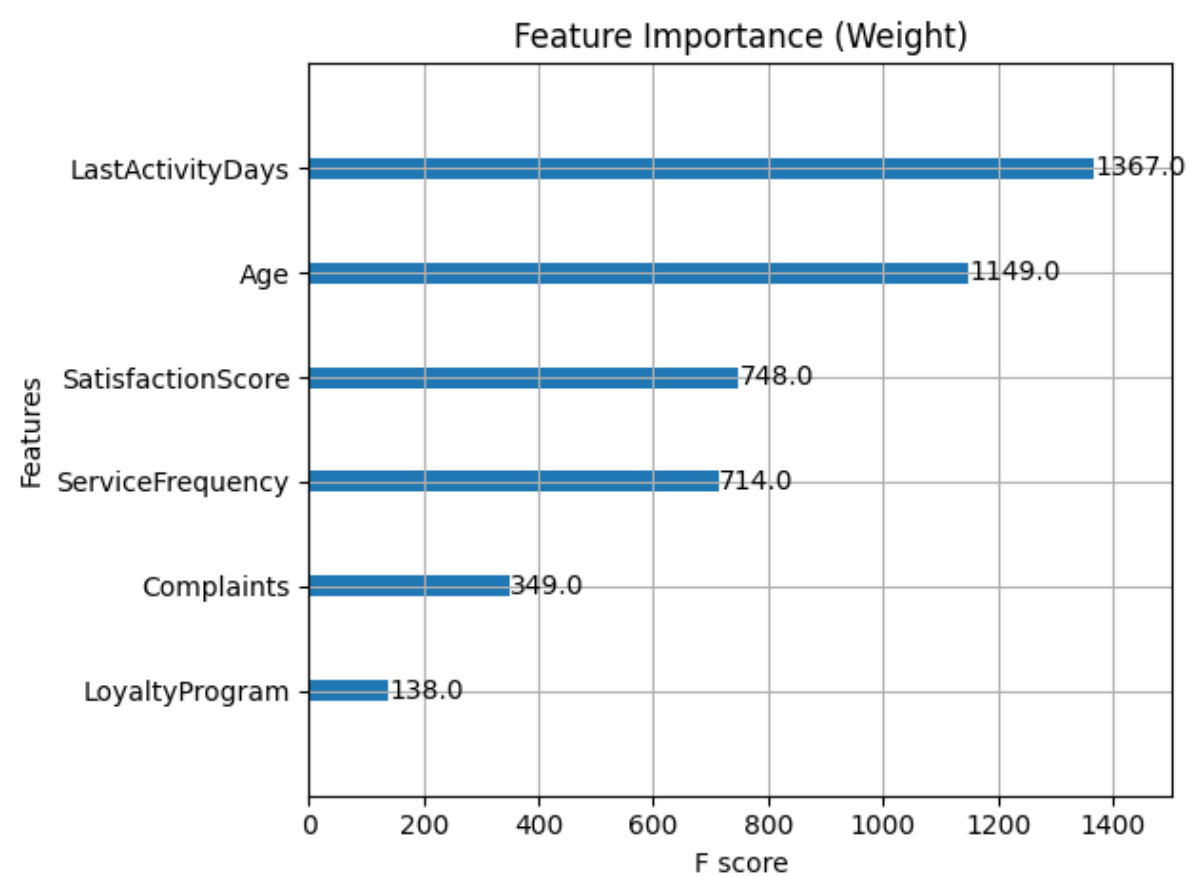


Figure 12: Feature importance Score

Recommendations for Business Decisions

After analyzing the visualizations, analytics, and predictive models, several actionable recommendations emerge to address the customer churn use case effectively. The primary focus should be on customer retention, as retaining existing customers is more cost-effective than acquiring new ones. Marketing campaigns via social media can engage high-risk customers, particularly those identified as having low spending and prolonged inactivity. Expanding the loyalty program to include a larger customer base can foster stronger brand commitment and incentivize continued engagement. Additionally, targeted campaigns tailored to specific customer segments can help reduce inactivity and improve overall engagement levels.

Customer satisfaction stands out as a critical area requiring immediate attention. Addressing customer complaints promptly and seeking feedback regularly are essential to building trust and

ensuring a positive customer experience. Predictive modeling is another key tool, enabling the identification of at-risk customers and allowing the organization to take proactive measures to prevent churn. Regular evaluations—conducted quarterly or biannually—are crucial to recalibrate strategies based on the latest data, as changes in customer behavior or market conditions may impact churn dynamics significantly. Also, leveraging segmentation insights to offer tailored promotions can drive engagement while maintaining cost efficiency. For example, personalized offers targeted at specific high-risk groups can maximize the impact of marketing efforts.

Forecasting: Flight and Hotel Booking

IN the next use case we focus our attention towards forecasting using time series data. Business case we aim to look into is forecasting the changing demands in the Hotel booking and the flight booking for our organisation. The dataset contains three columns: **Time** (monthly dates in YYYY-MM-DD format), **Hotel_Demand** (numerical values representing hotel demand), and **Flight_Demand** (numerical values for flight demand). The data is granular at the monthly level, consistently recorded at the end of each month. It exhibits a continuous and time-series-based nature, with values displaying seasonal fluctuations, such as higher demand during summer months (June to August) and lower demand during fall and winter months (September to February). Both **Hotel_Demand** and **Flight_Demand** show a general upward trend over the years, with seasonal peaks and troughs indicating their potential interdependence.

Figure 2.1 and **Figure 2.2** Visualizes the dataset that is generated synthetically for use in this particular usecase.

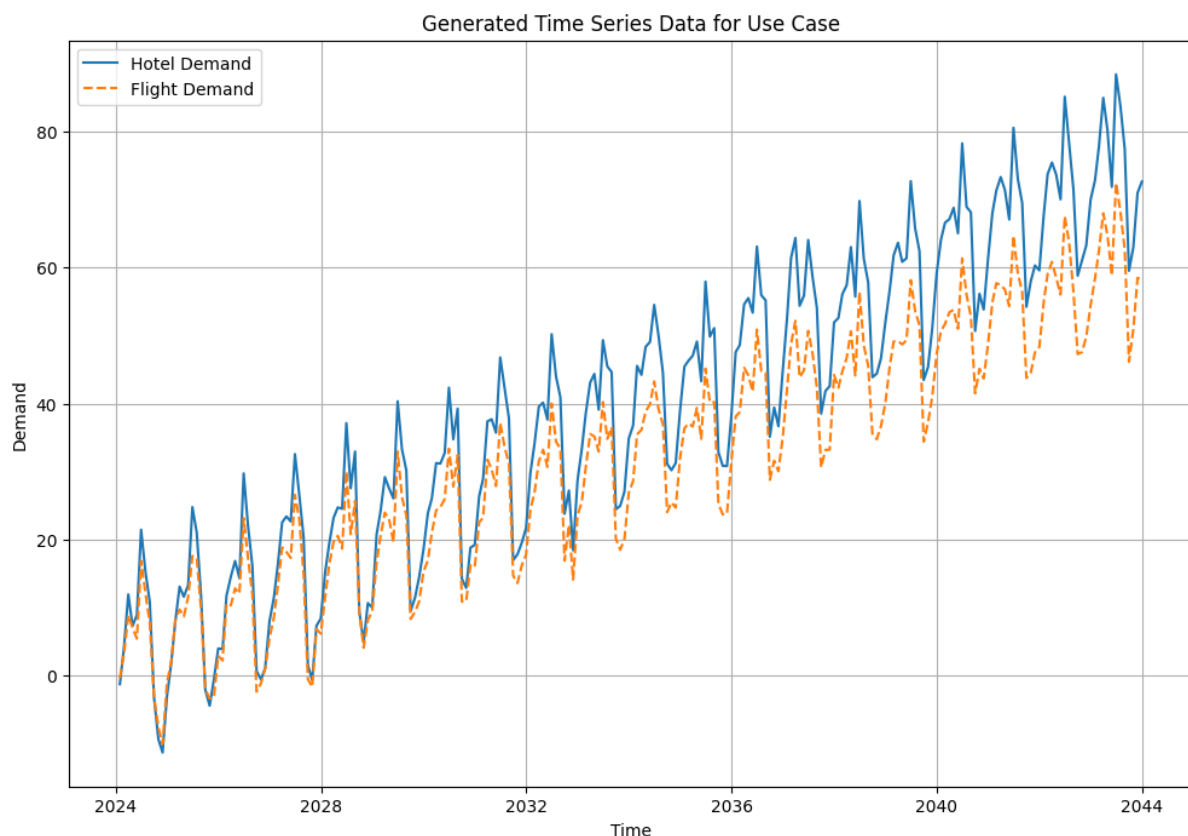


Figure 13: Visualizing time series data

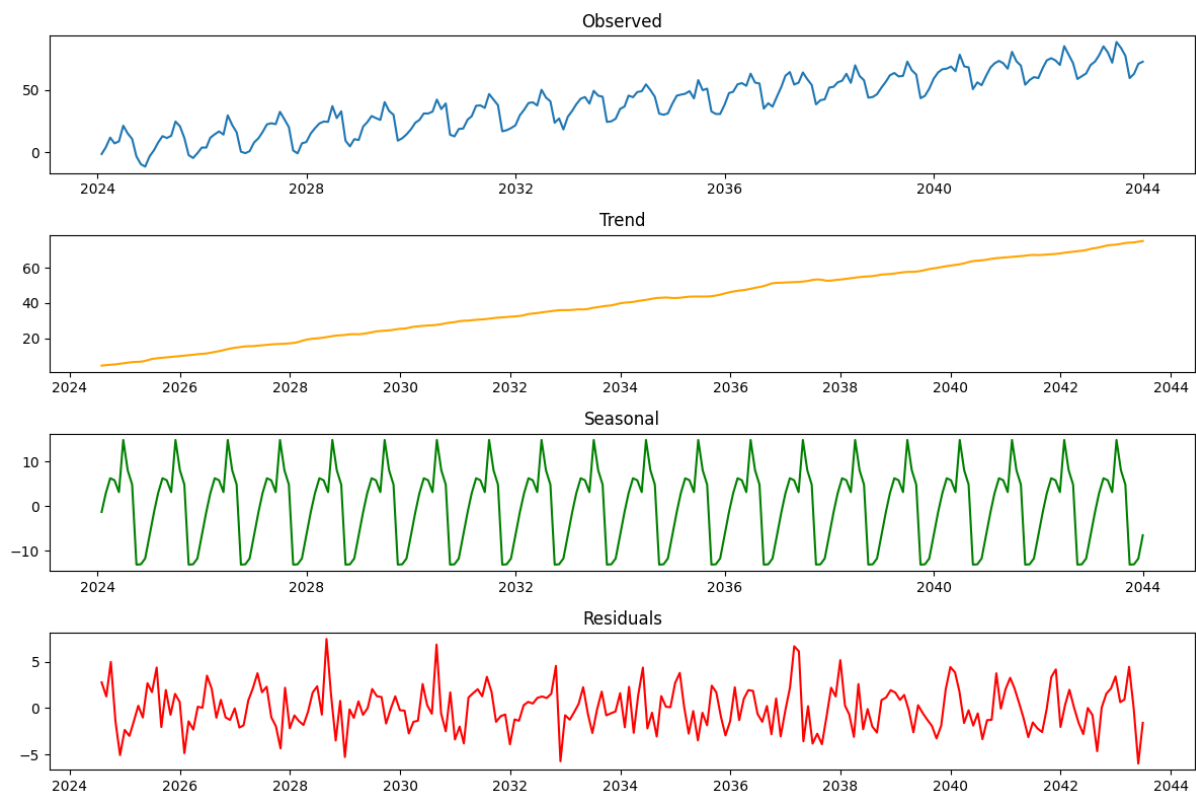


Figure 14: Breakdown of the data

As mentioned before there is an increasing trend in that is observed which is also visible from the graphs.

Our aim here is to ensure the organisation is prepared for the seasonality and trends that are observed over the years. Time series data is crucial for industries such as travel and tourism, Airlines, Hospitality and many others where helps in optimizing resources and enhance tourist experience. Adjust flight schedules, ticket pricing and arrangements of the aircraft in the case demand increases or deploy them to other partner companies incase they seccesfully forecast a decline in the demand. Similarly when talking about the hospitality arrangement of staff, room availability and offering of services as per fluctuating demand and season is important aspect that is look forward to by the decision makers.

Using forecasting and analysis techniques, what we aim to achive is to forecast and predict the rising or decline in the demands of to key feature. Hotel Demand and Flight booking.

As a first step we apply the Forecasting model to predict the demand for each of the feature we consider for this use case.

ARIMA

The ARIMA (AutoRegressive Integrated Moving Average) model is a statistical method used for time series forecasting by analyzing past data patterns. It combines three components: autoregression, integration, and moving average, which help in capturing temporal dependencies, removing trends to make the series stationary, and smoothing noise.

We aim to use ARIMA for forecasting hotel bookings and flight demand due to its ability to handle trends and short-term dependencies. The demand patterns in this particular usecase often exhibit non-stationarity and seasonality, which ARIMA addresses effectively through differencing and its AR and MA components. It excels in short- to medium-term forecasting, crucial for operational planning in the travel and hospitality industry, while requiring relatively smaller datasets compared to other machine learning models.

The figure 2.3 and Figure 2.4 shows the results of our application of ARIMA model for Htel and Flights.

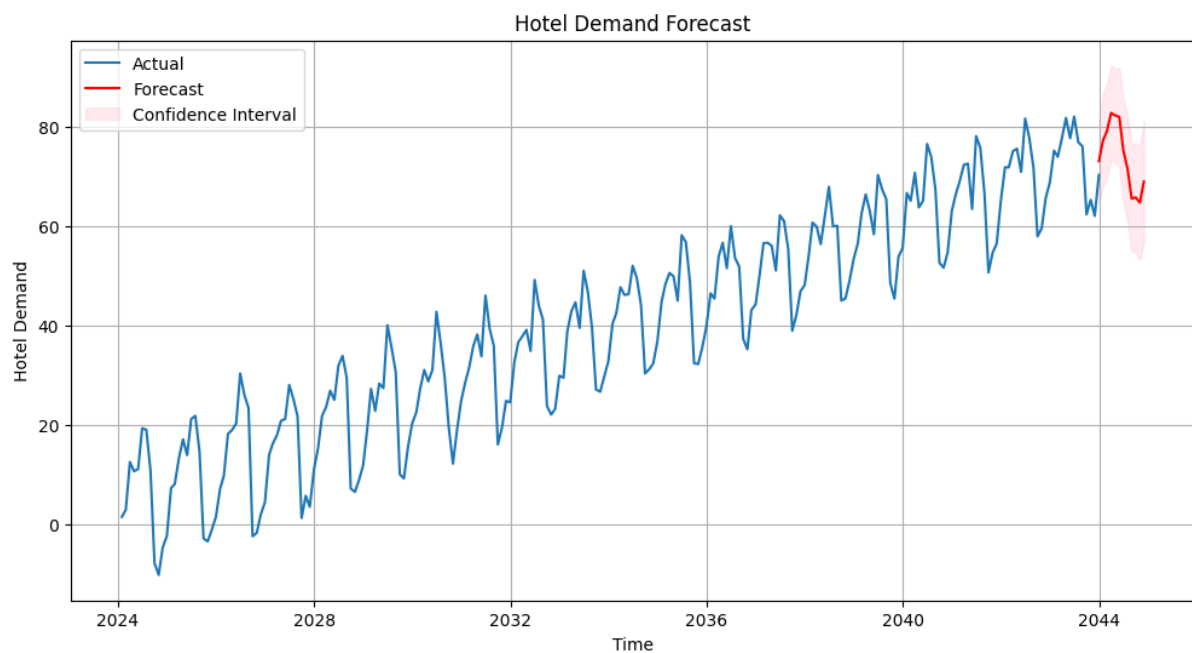


Figure 15: Forecasting Hotel demand

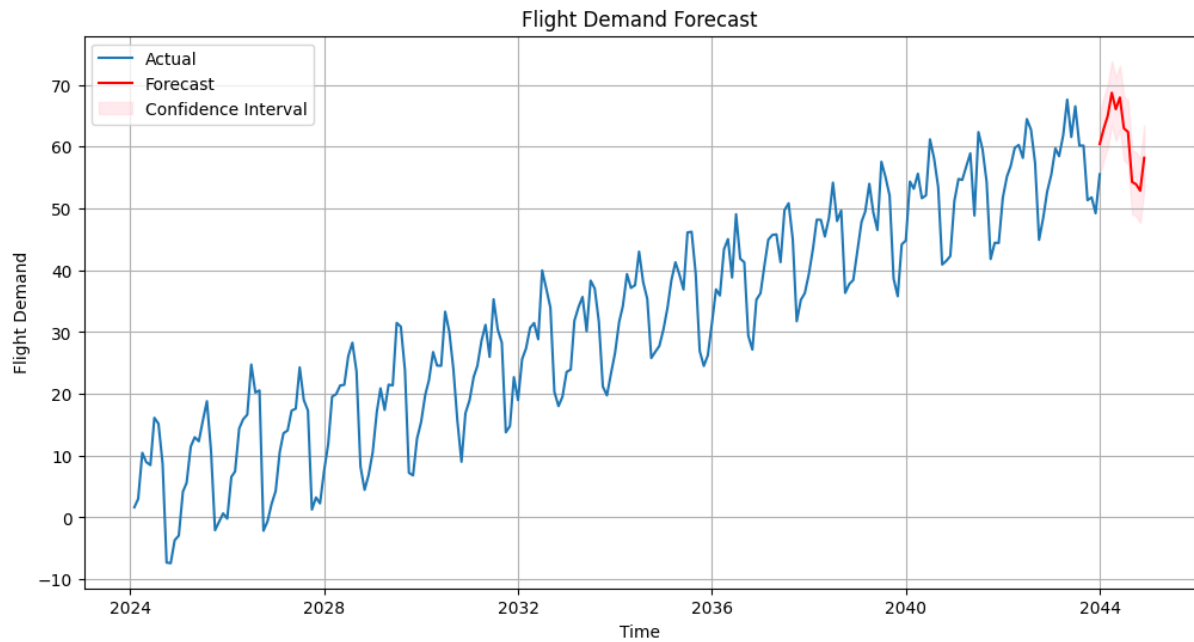


Figure 16: Forecasting Flight demand

The blue line on the graph represents the values in our dataset and the red-line represents the forecasted values for each of the feature. The shaded area around the forecasted value represents the confidence interval which is the range within which the actual future values are likely to fall. It accounts for the model's uncertainty and variability in predictions.

This accurate forecasting with high accuracy provide solid foundation to our oragnisation for decision making as it would be able to identify the demand patterens, seasonal fluctuations and potential growth opportunities. It helps hotel management adjust room availability and pricing strategies based on expected occupancy. And for the flights, knowing expected passenger loads allows for efficient fleet scheduling and route planning. Accurate demand forecasting also reduces the risk of overstaffing or understaffing, directly impacting labor costs and service quality which means considerable improvements in the profit.

Scenario

To understand the situation we created a virtual planning scenario which depicts the demand conditions such as unexpected surges or decline in the demands of Hotels and flights. This would help us prepare for uncertainties. Here we looked at an increase(50%) and decrease(25%) in the demand of Hotels. This particular demand can be seen by the graph in **figure 2.5** for the hotels and in a similar fashion for Flights in **figure 2.6**

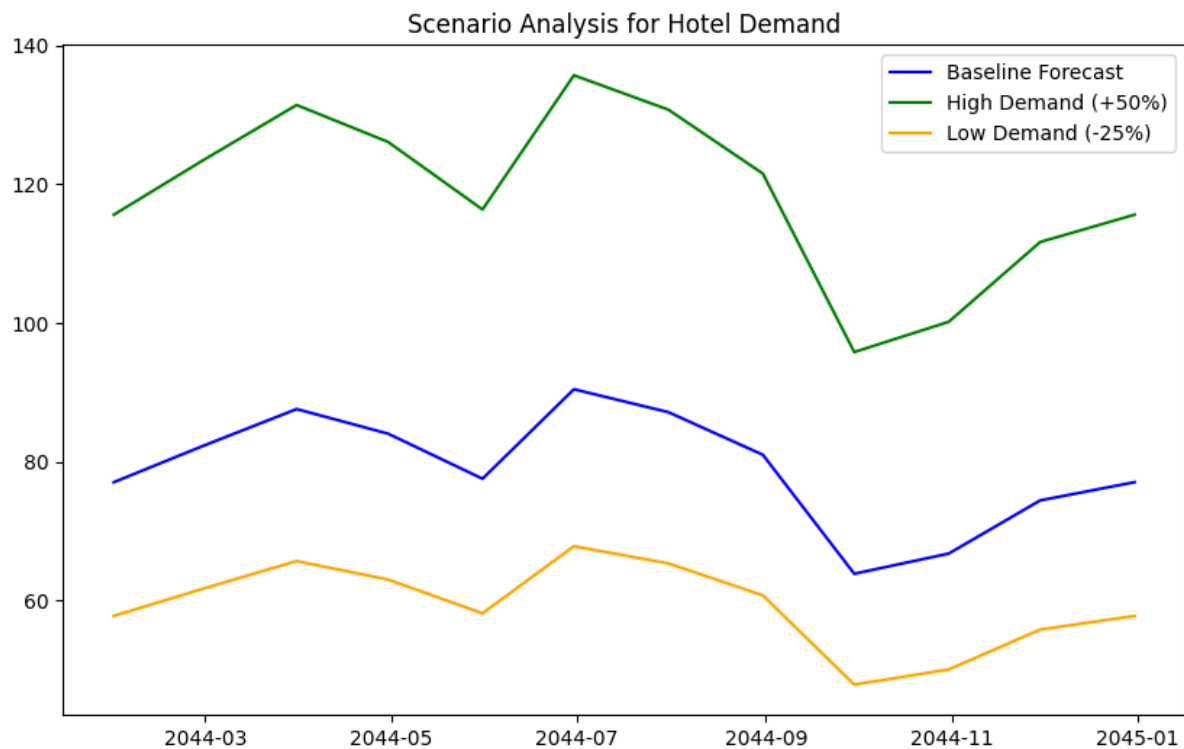


Figure 17: Scenario creation

Resource Allocation

Once faced with this situation next step to follow is the resource allocation which ensure that our company is caught offguard by the events like seasonal peaks, economic downturns, or sudden disruptions. For this we calculated the peak period from the scenario we created and this returned us the following values.

Table 3: Peak time Values

Peak time	Value
2044-02-29	81.036107
2044-03-31	87.177445
2044-04-30	90.481474
2044-06-30	84.079414
2044-07-31	87.626250
2044-08-31	82.264511

Considering the hotels accomodations for a resort can accommodate 20 people we can figure out the number of additional hotels that would be required as the forecasted demand (table below)

Table 4: Additional Hotels required

Peak time	Additional Hotel Required
2044-01-31	4.0
2044-02-29	5.0
2044-03-31	5.0
2044-04-30	5.0
2044-05-31	4.0
2044-06-30	5.0
2044-07-31	5.0
2044-08-31	5.0
2044-09-30	4.0
2044-10-31	4.0
2044-11-30	4.0
2044-12-31	4.0

This quotes us a figure about the number of Hotels (or per-person accommodations) that would be required to cater for the increased demand. Simulating higher-than-expected demand can also help prepare contingency plans, such as increasing staff availability, offering last-minute promotions, or leveraging partnerships with third-party providers. During periods of high demand, our organisation would ensure they have sufficient staff, inventory and other resources in place. Conversely, during low-demand periods, resources would be scaled down to reduce costs. It would greatly assist in hiring temporary staff, increasing housekeeping shifts, or expanding amenities during busy seasons. Similarly when we look into charter flights, it would help plan additional flights to meet peak travel demand.